



VEHICLE SPEED AND DENSITY ESTIMATION USING DEEP LEARNING ON CCTV FEEDS

Machine Learning Project

UNDER THE GUIDANCE
OF
JYOTHSNA DATTI

TEAM MEMBERS

01. Joshi Rushitha
2520030204

02. Manne Thanishka
2520030583

INTRODUCTION

With the increasing number of vehicles, traffic congestion and speeding have become major challenges. Traditional monitoring methods can be costly and time-consuming. This project uses existing CCTV cameras with deep learning to automatically monitor vehicles, estimate their speed, and analyze traffic density.

PROBLEM STATEMENT

Traditional traffic monitoring often depends on manual observation or additional sensors, making it difficult to monitor large areas continuously.

Our solution:

Use CCTV footage and deep learning to automatically detect vehicles, track their movement, estimate speed, and determine traffic density.

PROPOSED SYSTEM

The system takes CCTV footage as input and processes it using deep learning and computer vision techniques. Vehicles are detected and tracked across consecutive frames. Their movement is used to estimate speed, while the number of vehicles in a defined road area is used to calculate traffic density.



METHODOLOGY

CCTV Feed → Vehicle Detection →
Vehicle Tracking → Speed Estimation →
Vehicle Counting → Density Calculation →
Traffic Classification

YOLO can be used for fast and accurate
vehicle detection, while tracking
techniques follow vehicles between
consecutive frames

TECHNOLOGIES USED

- Python
- YOLO
- OpenCV
- Deep Learning
- Computer Vision
- CCTV Camera / Video Feed
- Deep Learning Frameworks

APPLICATIONS

01

- Smart city traffic monitoring
- Traffic control centers

02

- Road safety monitoring
- Traffic congestion analysis

03

- Intelligent transportation systems
- CCTV-based traffic management

Output

Dataset Loaded Successfully!

First 5 Rows:

	road_id	road_name	road_type	length_km	lanes	speed_limit_kmh	\
0	RD-0001	Jl. Segment 1	Jalan Kolektor	4.3	4	60	
1	RD-0002	Jl. Segment 2	Jalan Arteri	4.3	4	80	
2	RD-0003	Jl. Segment 3	Jalan Lokal	1.8	2	40	
3	RD-0004	Jl. Segment 4	Jalan Kolektor	1.8	4	60	
4	RD-0005	Jl. Segment 5	Jalan Tol	22.7	4	80	

	capacity_vehicles_per_hour	has_bus_lane	has_bike_lane	surface_condition	\
0	5384	0	0	Cukup	
1	6132	0	0	Baik	
2	978	0	0	Cukup	
3	4160	0	1	Rusak Berat	
4	6600	1	0	Baik	

	last_maintenance_date	latitude_start	longitude_start	latitude_end	\
0	2022-12-21	-6.173938	106.920654	-6.187271	
1	2022-11-17	-6.248396	106.768647	-6.246220	
2	2023-04-15	-6.267631	106.814963	-6.264098	
3	2022-05-17	-6.112216	106.954036	-6.122268	
4	2024-02-16	-6.349760	106.861835	-6.341074	

	longitude_end	district	toll_rate_rp	avg_daily_traffic
0	106.937674	PIK	0	10833
1	106.771962	Depok	0	6921
2	106.831007	Tangerang	0	1085
3	106.958261	Mangga Dua	0	6145
4	106.856648	Sudirman	7000	8023

Last 5 Rows:

	road_id	road_name	road_type	length_km	lanes	\
195	RD-0196	Jl. Segment 196	Jalan Kolektor	3.9	2	

Last 5 Rows:

		road_id	road_name	road_type	length_km	lanes	\
...	195	RD-0196	Jl. Segment 196	Jalan Kolektor	3.9	2	
	196	RD-0197	Jl. Segment 197	Jalan Arteri	3.3	6	
	197	RD-0198	Jl. Segment 198	Jalan Lingkungan	0.9	2	
	198	RD-0199	Jl. Segment 199	Jalan Kolektor	5.0	4	
	199	RD-0200	Jl. Segment 200	Jalan Tol	18.1	8	

	speed_limit_kmh	capacity_vehicles_per_hour	has_bus_lane	has_bike_lane	\
195	60	2198	0	1	
196	60	8130	1	0	
197	20	1386	0	0	
198	50	4776	0	0	
199	80	14744	1	0	

	surface_condition	last_maintenance_date	latitude_start	longitude_start	\
195	Cukup	2024-02-02	-6.233388	106.770735	
196	Baik	2024-03-17	-6.227477	106.949193	
197	Cukup	2023-09-02	-6.342497	106.741930	
198	Cukup	2022-06-13	-6.199783	106.954992	
199	Cukup	2024-01-01	-6.174337	106.778904	

	latitude_end	longitude_end	district	toll_rate_rp	avg_daily_traffic
195	-6.214167	106.752713	Menteng	0	2138
196	-6.208348	106.961205	Cibubur	0	6064
197	-6.348195	106.730341	Blok M	0	1446
198	-6.216635	106.956670	Cawang	0	6823
199	-6.157797	106.791170	Kemang	5000	16047

Dataset Shape:

(200, 18)

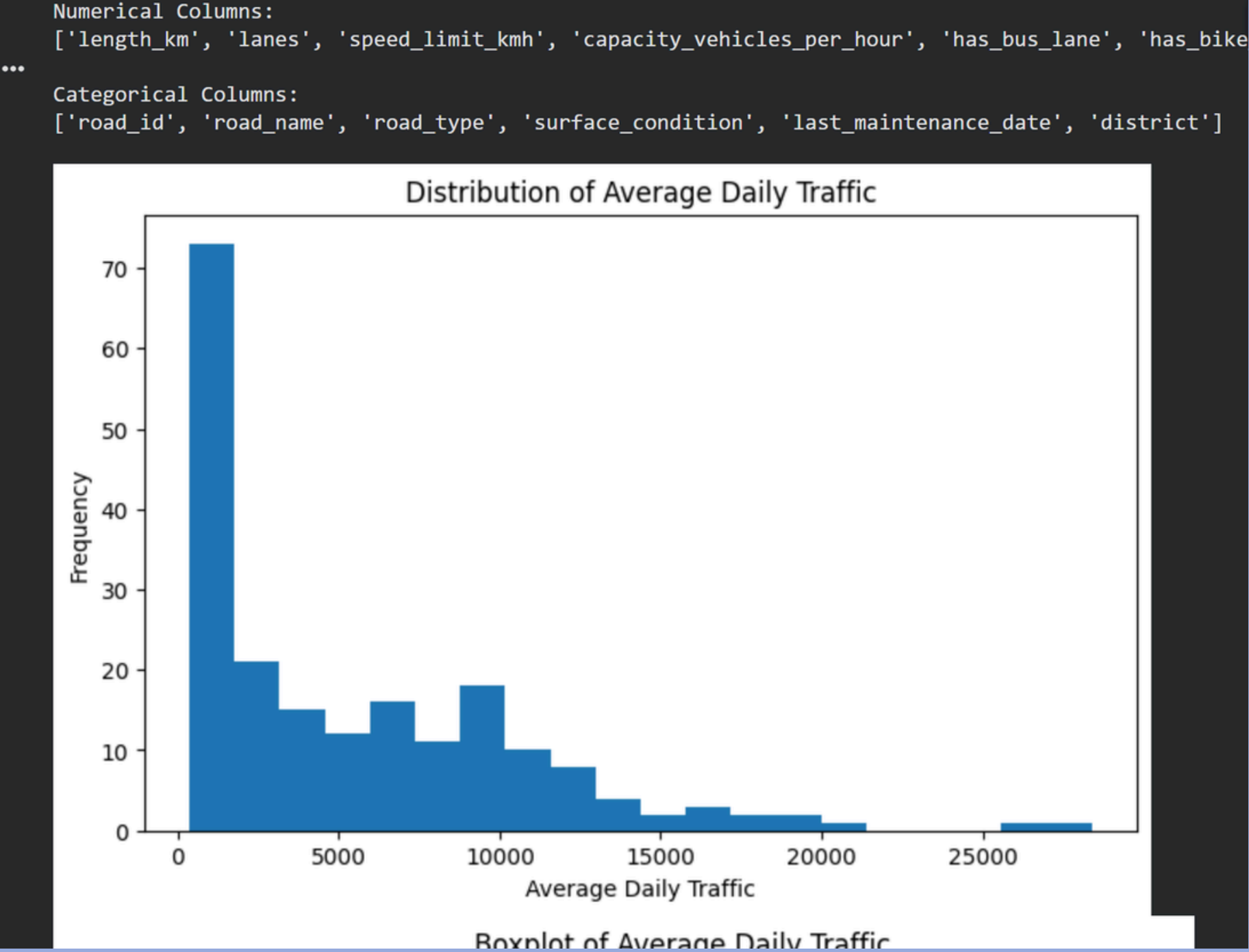
Number of Rows: 200

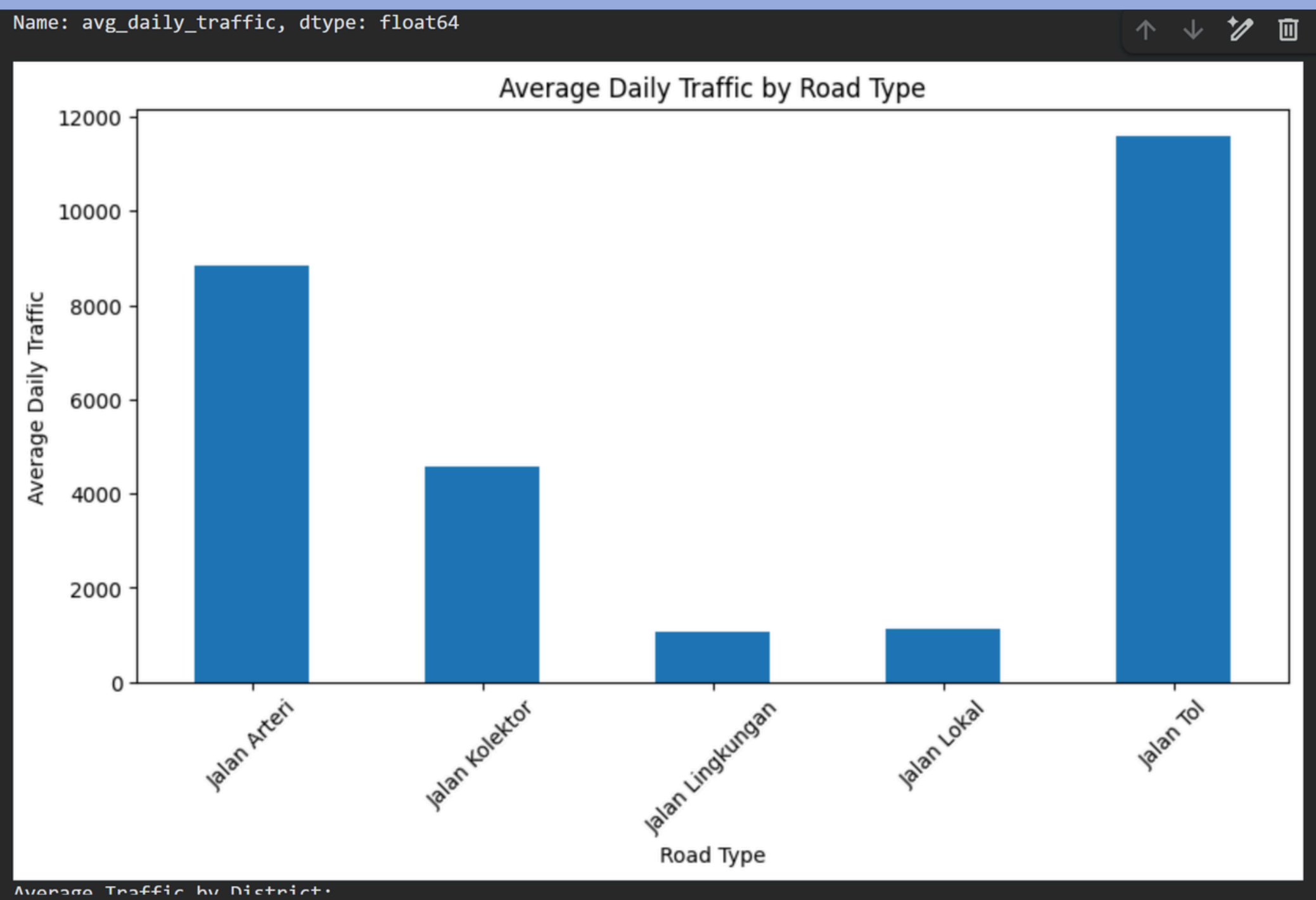
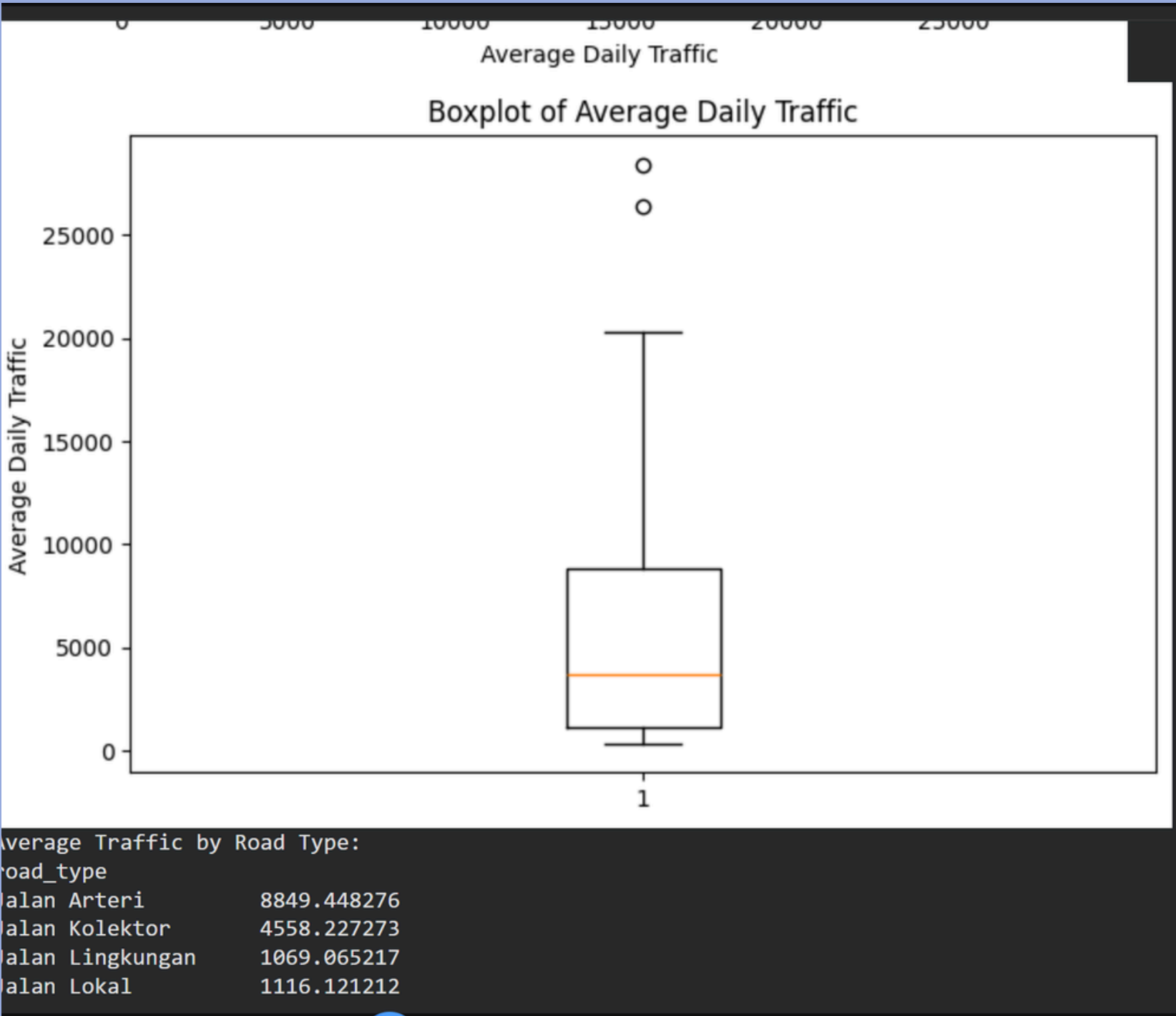
Number of Columns: 18


```
Column Names:
... ['road_id', 'road_name', 'road_type', 'length_km', 'lanes', 'speed_limit_kmh',

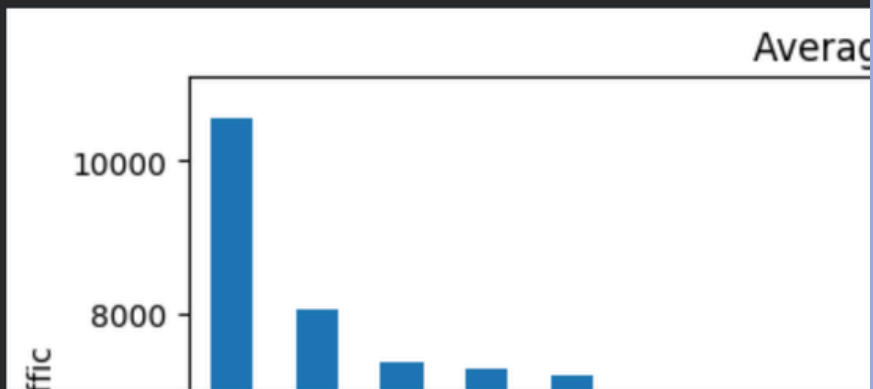
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   road_id                               200 non-null    object
1   road_name                             200 non-null    object
2   road_type                             200 non-null    object
3   length_km                             200 non-null    float64
4   lanes                                 200 non-null    int64
5   speed_limit_kmh                       200 non-null    int64
6   capacity_vehicles_per_hour            200 non-null    int64
7   has_bus_lane                          200 non-null    int64
8   has_bike_lane                        200 non-null    int64
9   surface_condition                     200 non-null    object
10  last_maintenance_date                  200 non-null    object
11  latitude_start                         200 non-null    float64
12  longitude_start                       200 non-null    float64
13  latitude_end                          200 non-null    float64
14  longitude_end                         200 non-null    float64
15  district                              200 non-null    object
16  toll_rate_rp                          200 non-null    int64
17  avg_daily_traffic                     200 non-null    int64
dtypes: float64(5), int64(7), object(6)
memory usage: 28.3+ KB

Statistical Summary:
      length_km    lanes  speed_limit_kmh  capacity_vehicles_per_hour \
count  200.00000  200.000000      200.000000      200.00000
mean     5.99450    3.335000       57.200000      4559.49000
```

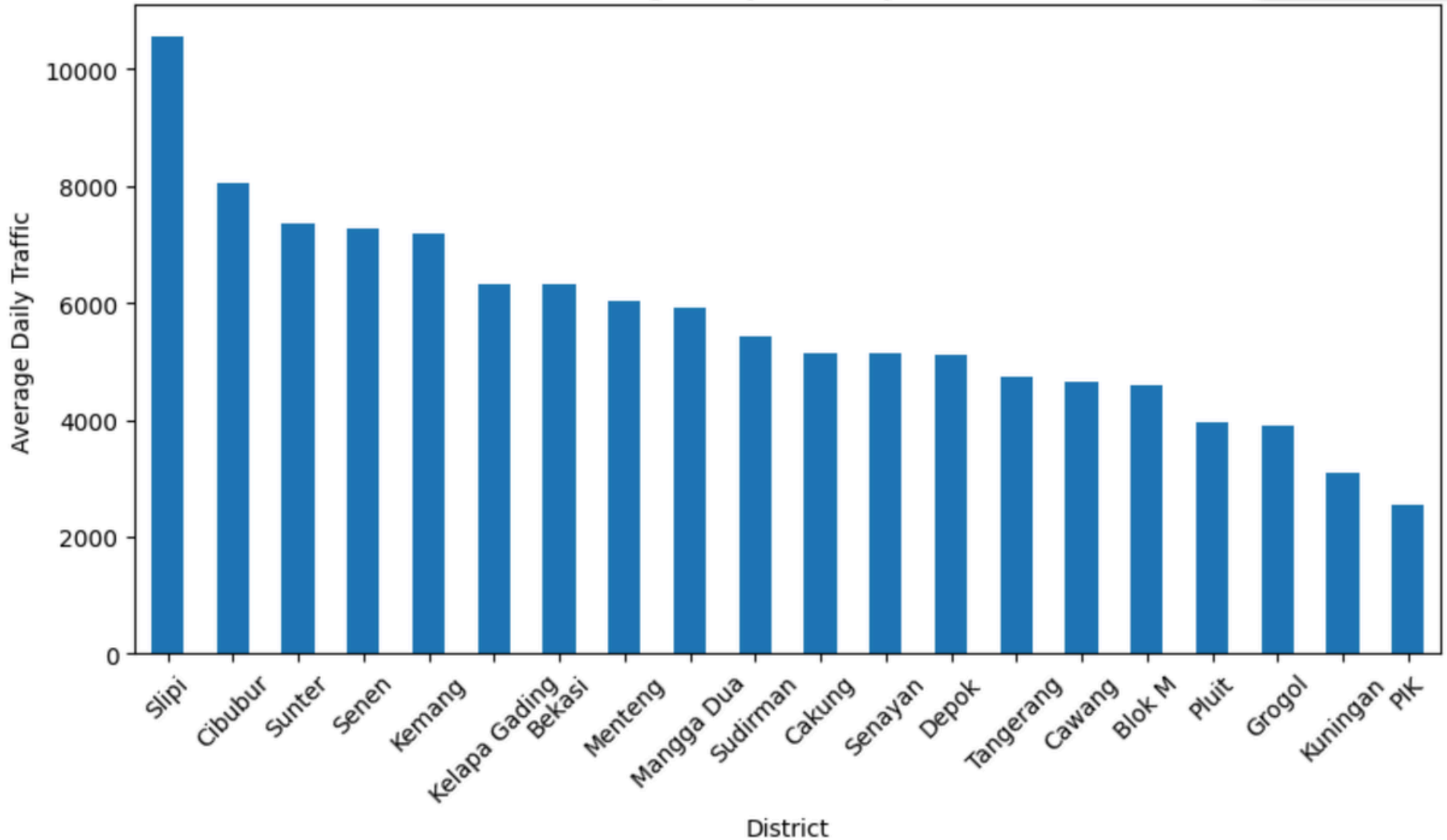




```
Average Traffic by District:
district
Slipi      10560.800000
Cibubur    8047.900000
Sunter     7357.714286
Senen      7272.357143
Kemang     7181.800000
Kelapa Gading 6314.666667
Bekasi     6313.000000
Menteng    6023.125000
Mangga Dua 5919.555556
Sudirman   5430.333333
Cakung     5148.166667
Senayan    5130.125000
Depok      5098.538462
Tangerang  4744.200000
Cawang     4647.900000
Blok M     4588.666667
Pluit      3972.500000
Grogol     3889.875000
Kuningan   3086.666667
PIK        2558.062500
Name: avg_daily_traffic, dtype: float64
```

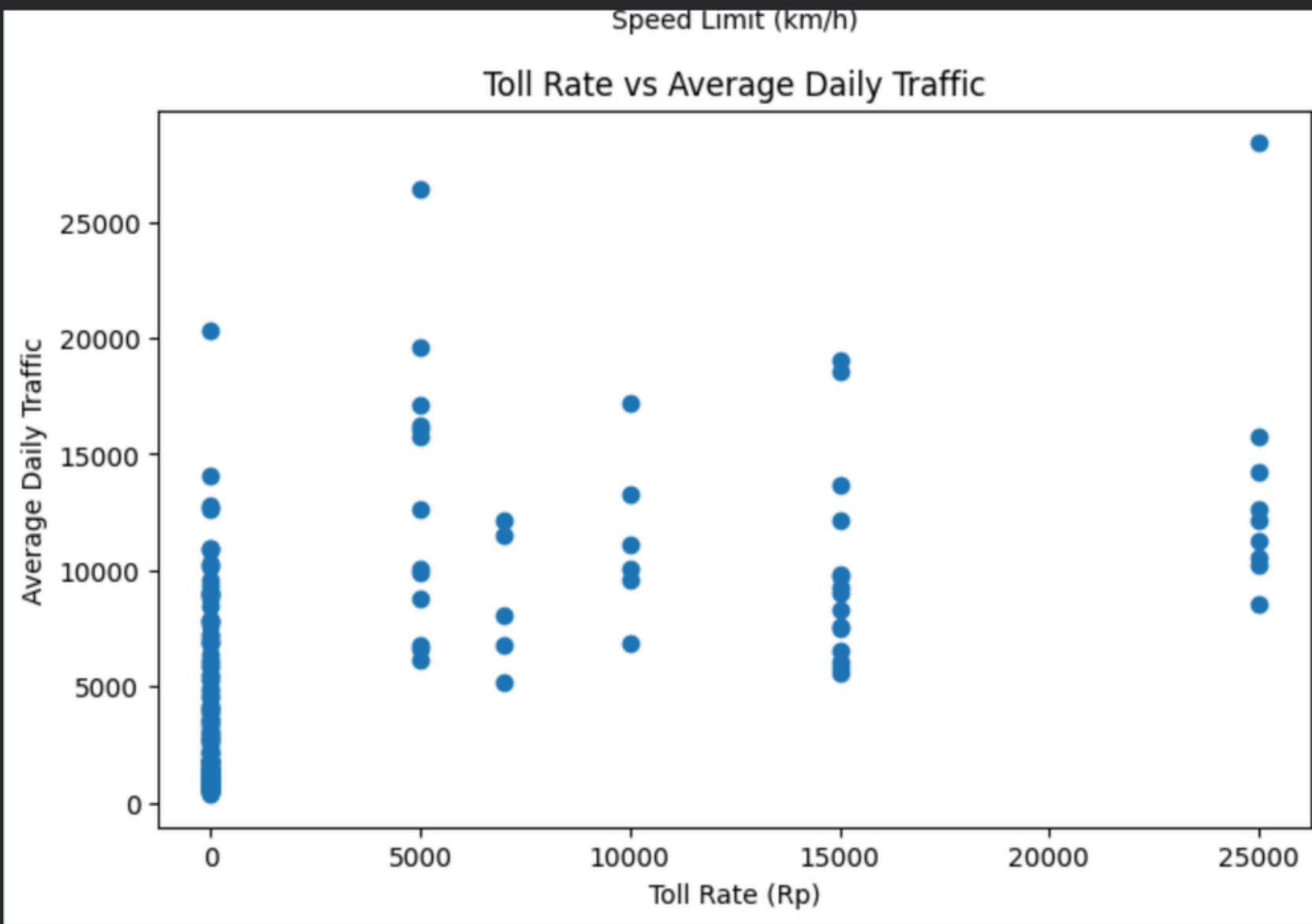


Average Daily Traffic by District



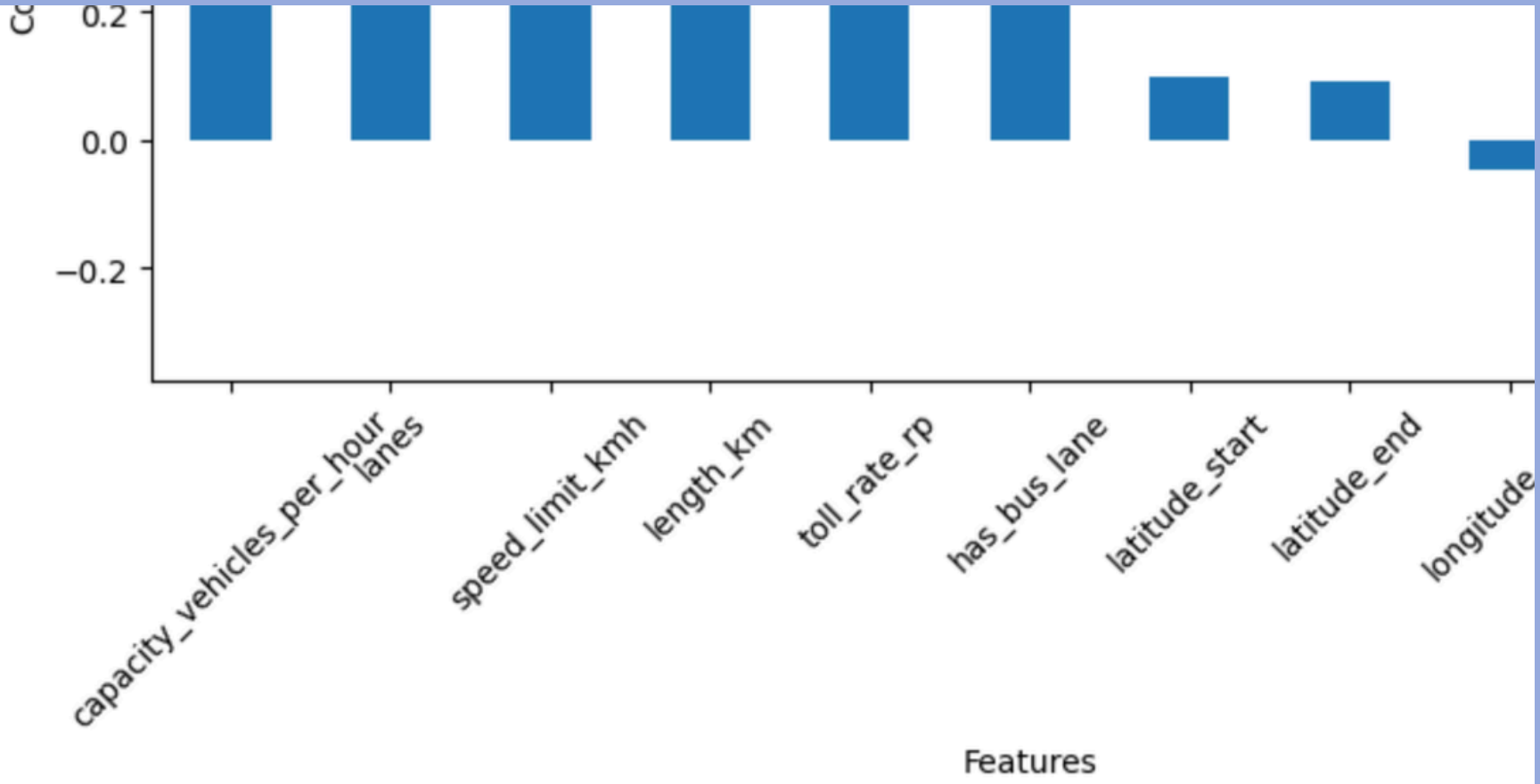
Number of Lanes vs Average Daily Traffic





Correlation Matrix:

	length_km	lanes	speed_limit_kmh	\
length_km	1.000000	0.743290	0.800769	
lanes	0.743290	1.000000	0.779259	
speed_limit_kmh	0.800769	0.779259	1.000000	
capacity_vehicles_per_hour	0.794835	0.972050	0.827924	



Number of Outliers in Average Daily Traffic:

FINAL EDA SUMMARY

Number of Samples: 200
Number of Features/Columns: 18
Missing Values: 0
Duplicate Rows: 0
Target Variable: avg_daily_traffic

Literature

Sno	Author Names	Title of the Paper	Year of Publication	Dataset Used	Methodology Used	Limitation of the Paper
1	Muhammad Umair Arif; Muhammad Umar Farooq; Rana Hammad Raza; Zain Ul Abideen Lodhi; Muhammad Abdur Rehman Hashmi	A Comprehensive Review of Vehicle Detection Techniques Under Varying Moving Cast Shadow Conditions Using Computer Vision and Deep Learning	2025	Highway I, II and III datasets; benchmark traffic video datasets	Review and comparison of computer-vision/deep-learning vehicle detection and shadow-removal techniques, including YOLO-based methods and shadow detection/removal.	It is a review rather than a new real-time system; performance still depends on dataset quality, lighting, shadows and camera conditions.
2	Kyung-Hee Kim; Tae-Ki An; Sunhee Kim	Estimating Invisible Passenger Count Using CCTV Footage: An Approach Combining Object Detection Models and Machine Learning	2025	Onboard railway CCTV footage	Two-stage approach using YOLOv8-L, Faster R-CNN, SSD-VGG16/ResNet50 followed by Random Forest, Gradient Boosting, SVR and XGBoost using region-wise features.	Occlusion and limited camera views remain difficult; results are mainly validated for railway passenger scenes.
3	Shraddha Bidwe; Geetanjali Kale; Ranjeet Bidwe	Traffic Monitoring System for Smart City Based on Traffic Density Estimation	2022	Highway CCTV video datasets; recorded road surveillance footage	YOLO/CFNN-based vehicle detection, classification, counting and traffic-density estimation with video processing.	Accuracy can decrease with low-quality CCTV, occlusion, changing illumination and complex traffic scenes.
4	Danesh Shokri; Christian Larouche; Saeid Homayouni	Real-Time Moving Vehicle Counting and Speed Estimation Toward Efficient Traffic Surveillance	2025	11+ million video frames from 511 highway cameras, NVIDIA AI City Challenge and Next Generation Simulation datasets	Spatial-Temporal Diagram generation, Gaussian Mixture Model refinement, RANSAC peak detection and traffic-parameter computation.	Performance can still be affected by severe occlusion and unusual camera/road configurations; calibration is important for speed estimation.
5	Zijian Hu; William H. K. Lam; S. C. Wong; Andy H. F. Chow; Wei Ma	Turning Traffic Surveillance Cameras into Intelligent Sensors for Traffic Density Estimation	2023	Traffic surveillance cameras from Hong Kong and Sacramento case studies	Camera calibration combined with deep-learning vehicle detection to estimate high-granularity traffic density.	Uncalibrated cameras, low frame rate, low resolution, limited annotations and complex road environments remain challenging.
6	Mallikarjun Anandhalli; Pavana Baligar; Santosh S. Saraf; Pooja Deepsir	Image Projection Method for Vehicle Speed Estimation Model in Video System	2022	CCTV traffic video footage	Image-plane to real-world-plane projection, camera/geometric calibration, vehicle detection and speed calculation.	Accuracy depends on correct camera calibration and geometric assumptions; unsuitable viewpoints can introduce speed errors.
7	Usha Mittal; Priyanka Chawla; Rajeev Tiwari	EnsembleNet: A Hybrid Approach for Vehicle Detection and Estimation of Traffic Density Based on Faster R-CNN and YOLO Models	2023	MB7500, KITTI and FLIR datasets	Data augmentation and image sharpening followed by a hybrid Faster R-CNN + YOLO majority-voting ensemble.	The ensemble has higher computational complexity and may require substantial processing resources for real-time deployment.
8	Victoria Miles; Francis Gurr; Stefano Giani	Camera-Based System for the Automatic Detection of Vehicle Axle Count and Speed Using Convolutional Neural Networks	2022	Roadside video-camera recordings	Two YOLOv3 CNN models for axle detection and speed measurement, with Kalman-filter vehicle tracking and road-marking calibration.	Kalman tracking struggles during long occlusions; speed calibration introduces error when road markings do not follow standard specifications.
9	Jazem Mutared Alanazi	Effective Machine-Learning Based Traffic Surveillance Moving Vehicle Detection	2023	Traffic surveillance video datasets	Block-level DCT feature extraction combined with edge/texture features using DRLTP for moving-vehicle detection.	Handcrafted transform-domain features may be less adaptable than modern deep models to highly varied scenes and camera conditions.
10	G. Priyanka; J. Senthil Kumar; S. T. Veena	Deep Learning Based Video Surveillance for Predicting Vehicle Density in Real Time Scenario	2023	UCSD traffic dataset	Multi-Scale CNN (MSCNN) and Single-Scale CNN (SSCNN) for extracting vehicle features and classifying traffic density as low,	Evaluation is based mainly on the UCSD dataset; generalization to different camera angles, weather and dense urban scenes

No.	B		C		D		E		F		G	
	▼	Author Names	▼	Title of the Paper	▼	Year of Publication	▼	Dataset Used	▼	Methodology Used	▼	Limitation of the Paper
11		Dini Oktarina Dwi Handayani; Teddy Mantoro; Firman Arifman; Ben Rahman; Media Anugerah Ayu; Rahmadya Trias Handayanto		A Canonical Computational Intelligence Framework for Adaptive Traffic Volume and Density Estimation From Legacy Analog Surveillance Systems		2026		Real analog toll-road CCTV footage		ACI-TA framework combining restoration, vehicle perception, identity reasoning, spatial logic and statistical traffic analytics for degraded analog video.		Designed around degraded analog surveillance; broader validation across countries, camera types and traffic environments is still required.
12		Oriol Alás; Ricard Cervera; Robert Sanfeliu		A Scalable Real-Time Multi-Camera Vehicle Tracking System for Urban Environments		2026		Public multi-camera benchmarks and a real-world logistics hub		Lightweight YOLO-based detection, transformer appearance embeddings, topology/travel-time constrained association and edge-fog deployment.		Severe occlusion, viewpoint changes, camera synchronization and cross-camera re-identification remain challenging.
13		Enkusilasie Teegne Alemu; Gizatie Desalegn Taye		Development of a Cost-Effective Data Collection System for Monitoring Urban Traffic Using Automated Intelligent Image Processing and Machine Vision		2026		3,599 traffic-video images including Hungarian license-plate images		Integral Image representation, AdaBoost feature selection, Haar-cascade classification, ROI processing and OpenCV/Python.		The evaluation is focused on Hungarian license plates; performance may reduce for other plate designs, environment and severe image degradation.
14		Chelsie D. Bajamunde; Mark Harold B. Bien; Thrisia A. Sarcilla; Referendo D. Soriano		Performance Comparison of Object Detection Models for Traffic Density Estimation Using CCTV Footages		2026		CCTV footage from Elias Angeles St., Naga City; 320×320 processed images		Adaptive background subtraction and morphological opening followed by comparison of Faster R-CNN, YOLOv3 and SSD for five vehicle classes.		Results are location-specific and depend on CCTV quality; density estimation based on vehicle-size assumptions may not generalize to all roads.
15		Paganelli Mariano Giuseppe		Integrated Passenger Flow Analysis and Street-Level Classification for Public Transport Management using Deep Learning and IoT		2025		Public-transport onboard video streams, GPS and street-level observations		YOLO-based passenger counting and external object detection integrated with GPS/geospatial analysis, IoT and cloud processing.		The system requires multiple sensors/cameras and cloud infrastructure; performance can be affected by occlusion, camera placement and connectivity.
16		Siyuan Tang		Using Unmanned Aerial Vehicles (UAV) and Closed-Circuit Television (CCTV) Video for Multi-Object Multi-Camera Vehicle Tracking in Traffic Monitoring		2025		UAV and CCTV traffic-monitoring video		MOMCMVT framework combining UAV and CCTV views with multi-object tracking, CSRT tracking and cross-camera/multi-view information.		Synchronizing heterogeneous UAV/CCTV views and maintaining identity under occlusion and viewpoint changes remain difficult.
17		Jiyasmin Akter Sinthiya; Md Musfiqur Rahman; Rifat Ibne Yousuf; Md. Tashin Parvez; Md. Azizul Haque Noman; Raiyan Rahaman		AutoTrack: A Deep Learning Approach on a Novel Dataset for Vehicle Detection and Route Tracking from CCTV Images		2025		Novel CCTV vehicle dataset collected for the study		YOLOv5 for vehicle/color detection, YOLOv8 for license-plate recognition, time-stamped CCTV path reconstruction and geospatial mapping.		Performance is affected by poor CCTV quality, sparse camera coverage and occlusion; validation is based on a relatively specific surveillance scenario.
18		Author(s) reported in the uploaded SSRN paper		Real-Time Traffic Density Estimation Using YOLO-Based Deep Learning Models on Urban Surveillance Datasets		2026		Urban surveillance datasets		YOLO-based deep-learning vehicle detection and traffic-density estimation for real-time urban monitoring.		Performance degrades with occlusion, varying illumination, dense congestion, limited annotations and computational constraints.
19		Ade Kurniawan		Real-Time Smart Traffic Monitoring System Using YOLOv12 and ByteTrack: An Integrated Approach for Vehicle Detection, Counting, Speed Estimation, and Incident Detection		2026		Traffic surveillance video datasets		YOLOv12 for vehicle detection combined with ByteTrack for multi-object tracking, counting, speed estimation and incident detection.		Real-world deployment may be constrained by computational resources, camera calibration, occlusion and variations in traffic/weather conditions.
20		Dini Oktarina Dwi Handayani; Teddy Mantoro; Firman Arifman; Ben Rahman; Media Anugerah Ayu; Rahmadya Trias Handayanto		A Canonical Computational Intelligence Framework for Adaptive Traffic Volume and Density Estimation From Legacy Analog Surveillance Systems (duplicate copy in second archive)		2026		Real analog toll-road CCTV footage		ACI-TA computational-intelligence pipeline for degraded analog CCTV, vehicle counting and traffic-density estimation.		The second uploaded copy is the same research paper; independent validation on additional datasets and camera configurations is still needed.

CONCLUSIONS

This project demonstrates how deep learning and computer vision can be combined with CCTV feeds for automated traffic monitoring. By detecting and tracking vehicles, the system can estimate speed and traffic density, providing useful information for safer and more efficient traffic management.



THANK YOU
VERY MUCH!

